

2025 Semester 1

01211373 Machine Learning and Programming for Industry

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Lecture 3

▼ Linear and Logistic Regression

AI's explanation on the term "regression"

- "The term "regression" in statistics comes from Sir Francis Galton's observations about heredity, specifically the tendency of offspring to regress towards the mean of a population. In the late 1800s, Galton studied the relationship between parents' and children's heights and noticed that exceptionally tall or short parents tended to have children whose heights were closer to the average height of the population. This "regression towards the mean" is the origin of the term, which later evolved into the broader statistical concept of regression analysis."

Reference videos for linear regression

- Stanford CS229 ML course (Andrew NG 2018), Lecture 2 : Linear Regression and Gradient Descent
https://youtu.be/4b4MUYve_U8?si=qa3iBTO2R-abQViu
- MIT: Machine Learning 6.036, Lecture 5: Regression (Fall 2020)
<https://youtu.be/OpvrwhWEUmY?si=bS8VZYsrZMqvnu>
II

Simple Linear Regression

(see lecture3.ipynb for more)

- In general, find $h_\theta(x)$ that maps x in $\mathbb{R}^d$ to y in $\mathbb{R}$ (d is dimension of feature vector)
- $h_\theta(x)$ is called a hypothesis or model
- In linear regression literature, $\theta_i, i = 1, 2, \dots$ is often used in place of w_i and θ_0 for b .
- First consider simple case $h_{\theta(x)} = \theta_0 + \theta_1 x_1$
- Define intercept term $x_0 = 1$ so that
 - $h_\theta(x) = \sum_{i=0}^d \theta_i x_i = \theta^T x$
- Define cost function

$$J(\theta) = \frac{1}{2} \sum_{i=1}^n (h_\theta(x^{(i)}) - y^{(i)})^2$$

LMS algorithm

Objective : Choose θ to minimize $J(\theta)$

Gradient descent

- start with some initial guess for θ
- Update θ to minimize $J(\theta)$

$$\theta_j := \theta_j - \alpha \frac{\partial}{\partial \theta_j} J(\theta), \quad j = 0, \dots, d$$

Here α is called *learning rate*.

It is shown in [2] that

$$\frac{\partial}{\partial \theta_j} J(\theta) = (h_{\theta}(x) - y)x_j$$

Hence, the LMS update rule for single training example becomes

$$\theta_j := \theta_j + \alpha (y^{(i)} - h_{\theta}(x^{(i)}))x_j^{(i)}$$

a.k.a **Widrow-Hoff** learning rule.

LMS Algorithm

For the whole data set, the algorithm becomes

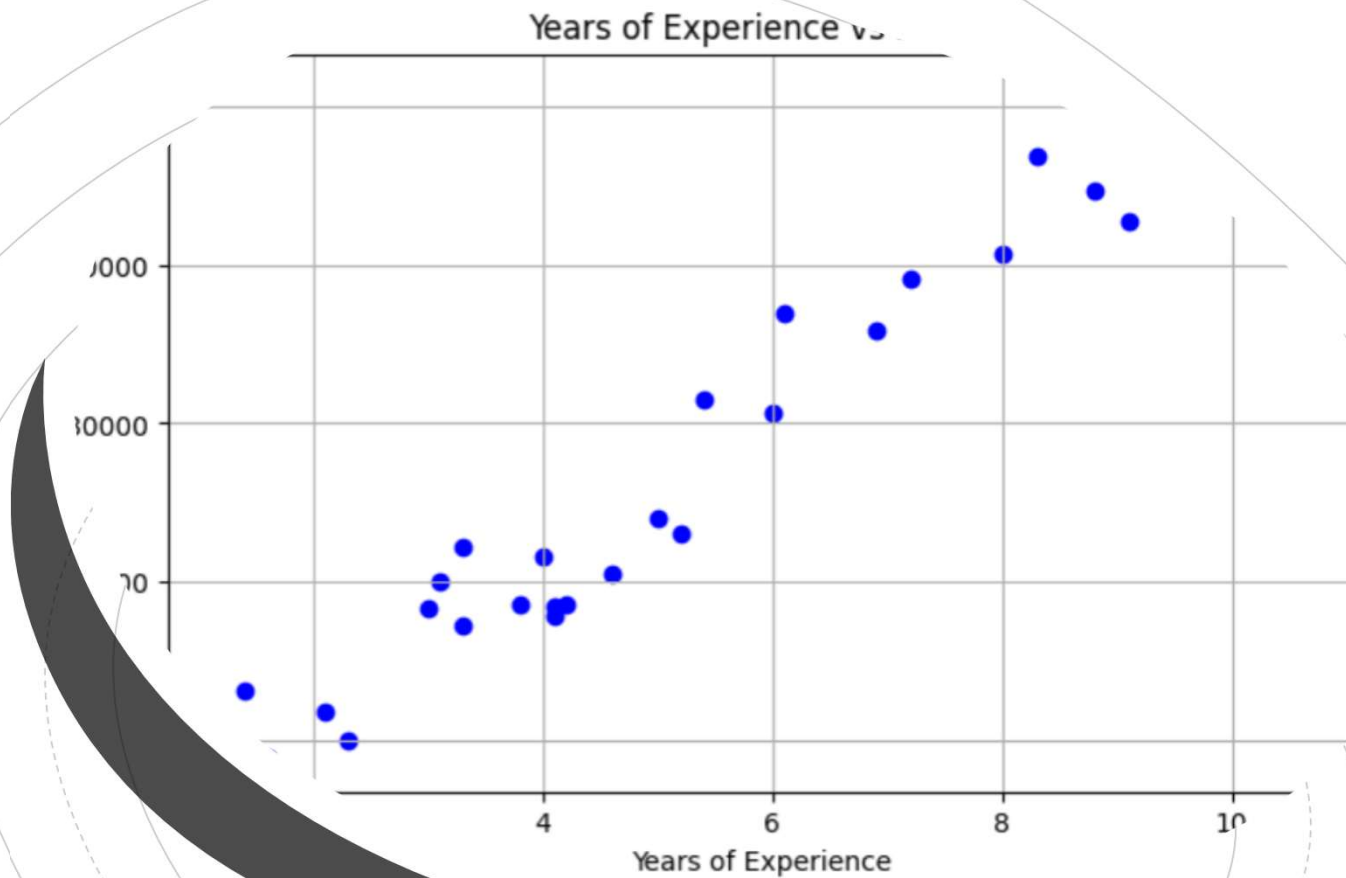
Repeat until convergence {

$$\theta_j := \theta_j + \alpha \sum_{i=1}^n (y^{(i)} - h_{\theta}(x^{(i)})) x_j^{(i)}, \quad j = 0, \dots, d$$

}

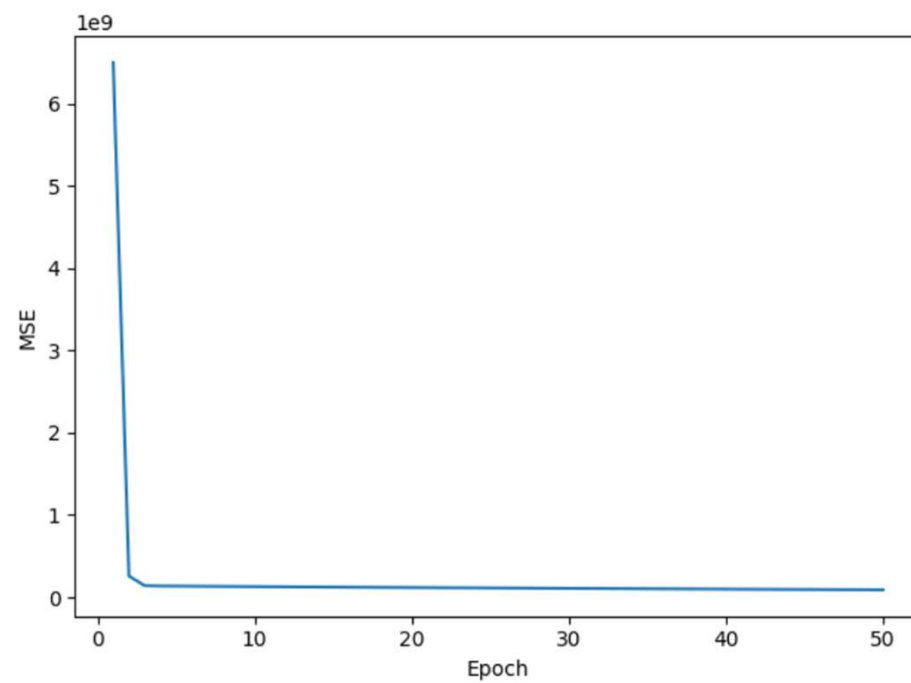
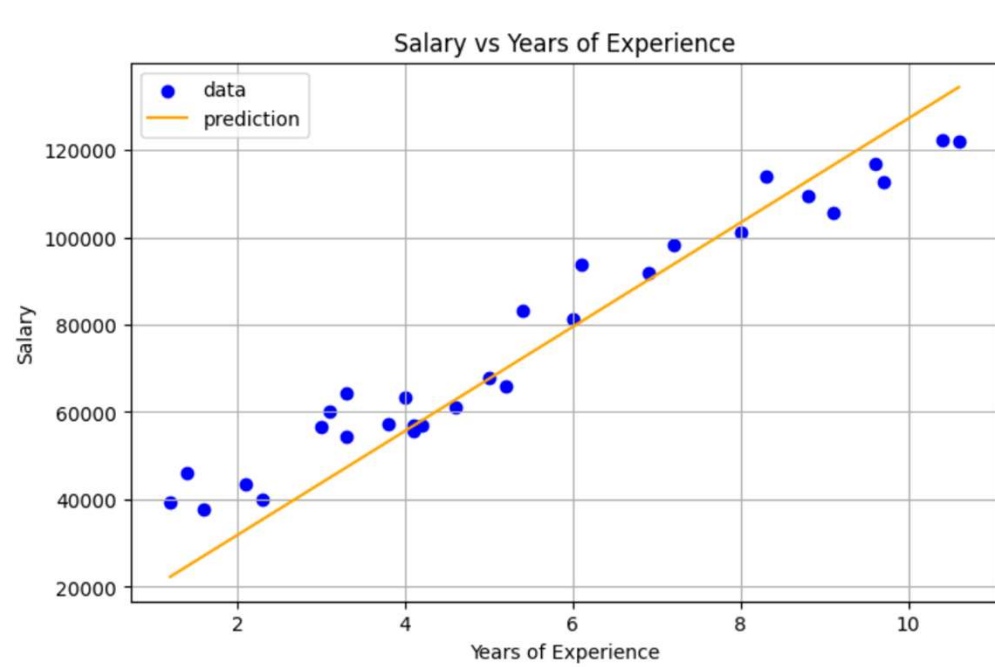
Or, compactly with θ in vector form

$$\theta := \theta + \alpha \sum_{i=1}^n (y^{(i)} - h_{\theta}(x^{(i)})) x^{(i)}$$



Simple salary
dataset
from
kaggle

<https://www.kaggle.com/datasets/abhishek14398/salary-dataset-simple-linear-regression/data>



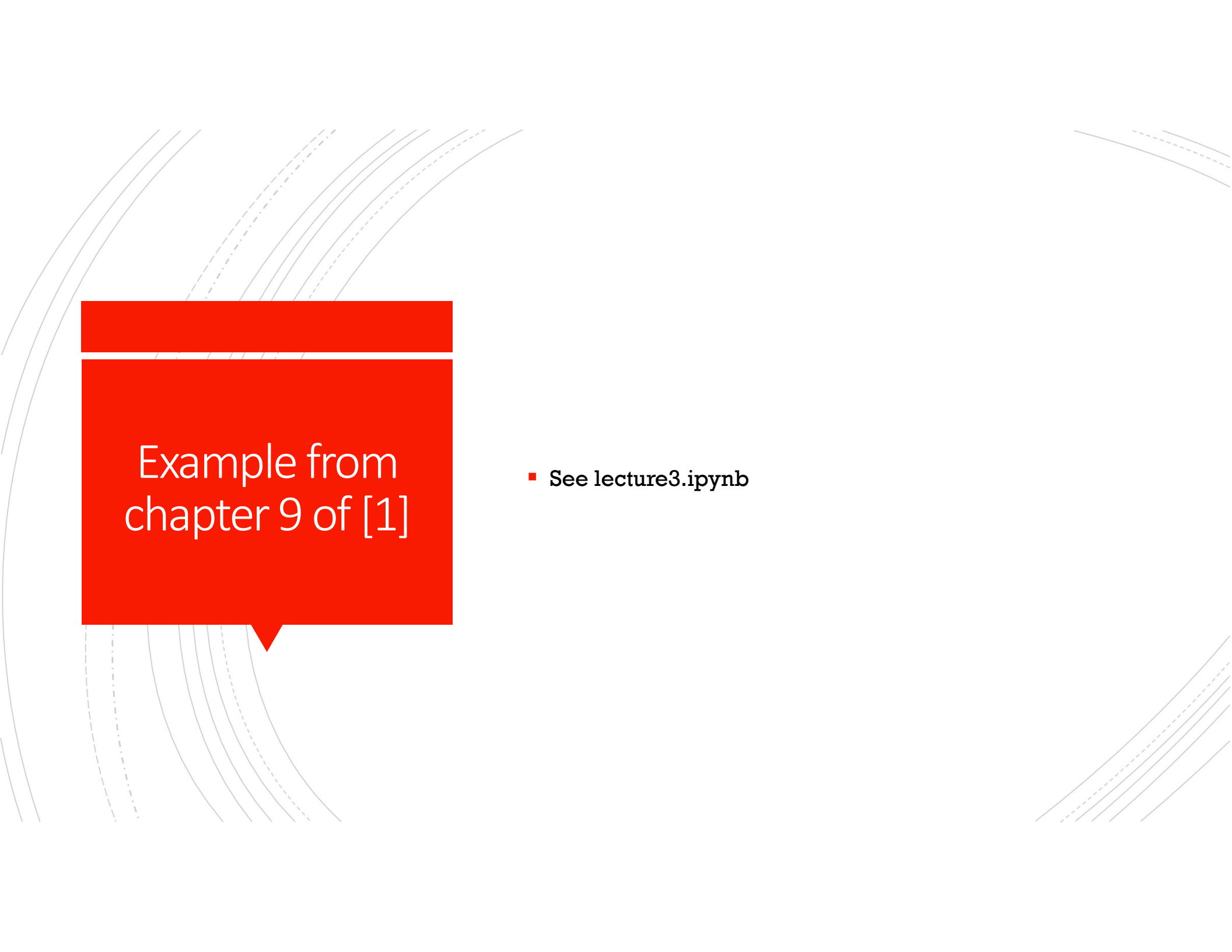
Result from
linear regression

Normal equation

- The value of θ that minimizes $J(\theta)$ is given in closed form by

$$\theta = (X^T X)^{-1} X^T y$$



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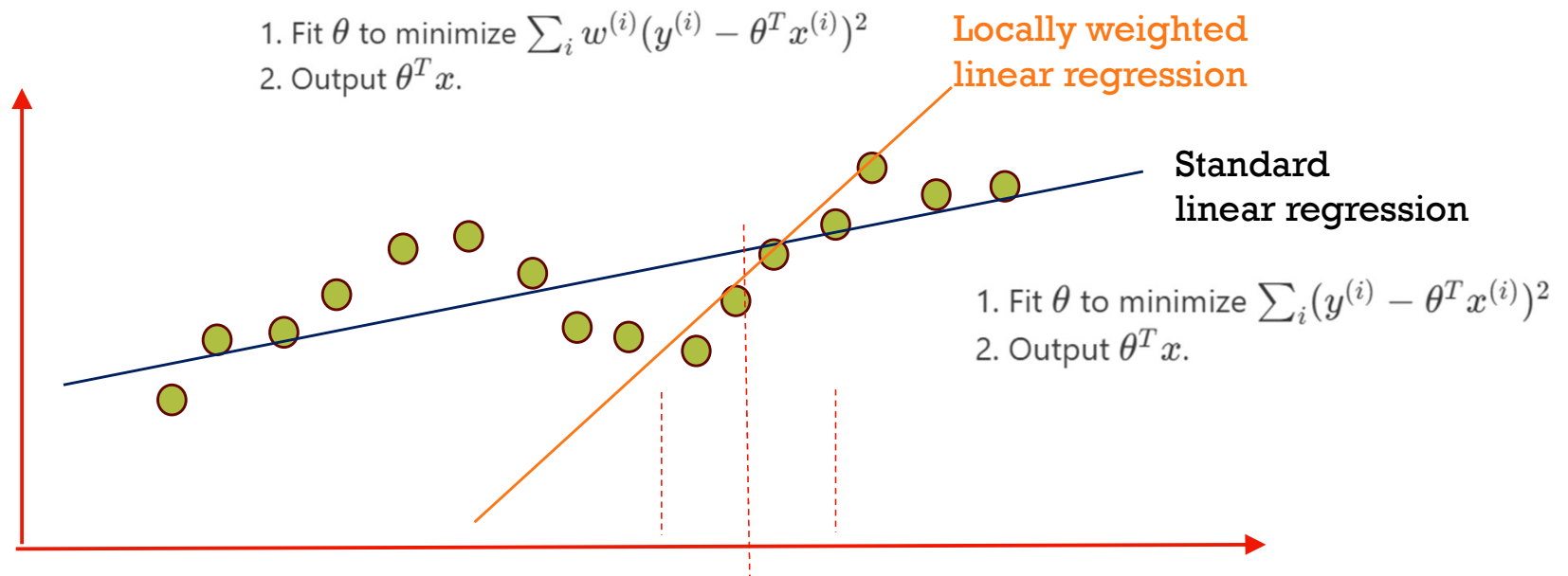
Example from
chapter 9 of [1]

- See `lecture3.ipynb`

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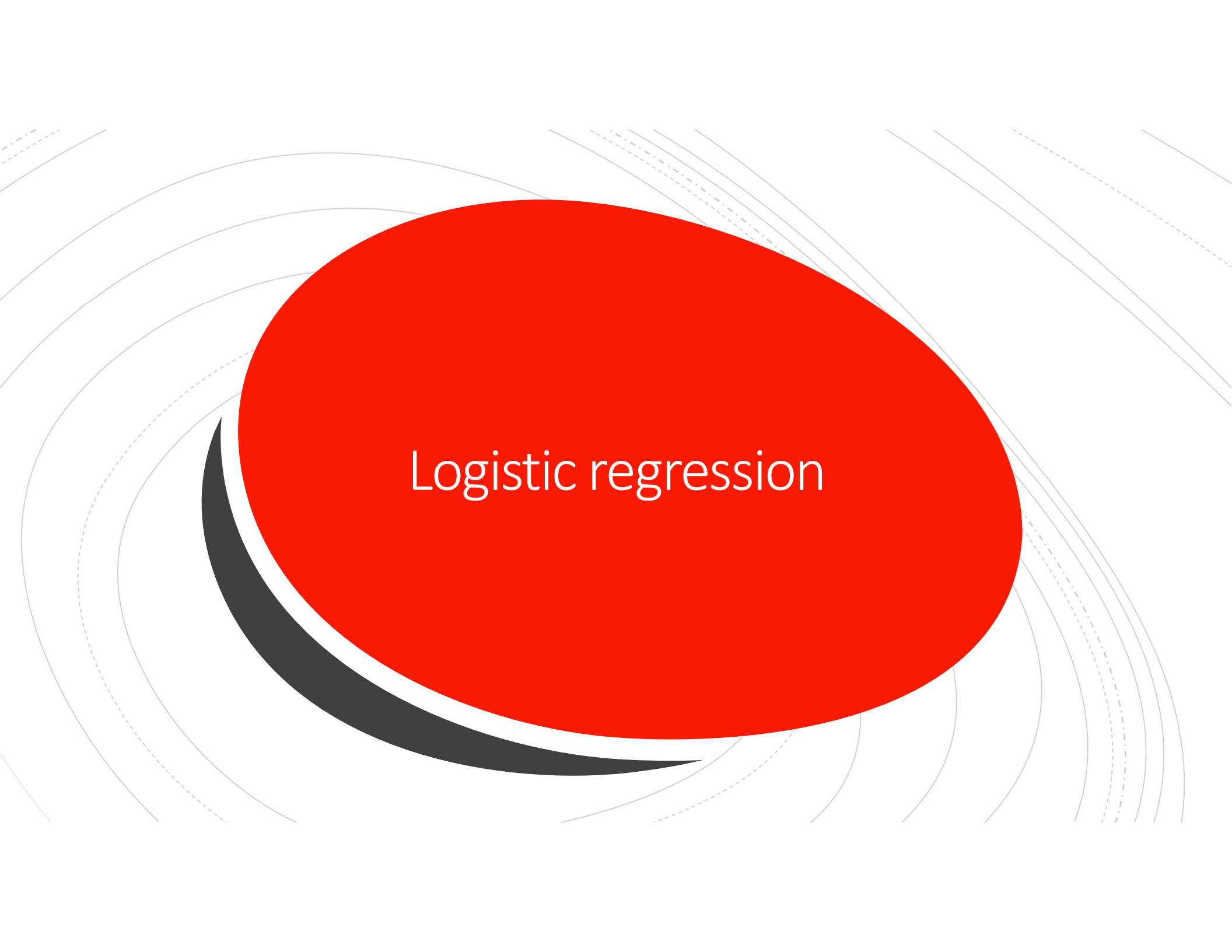
Optional reading for linear regression from [2]

- Probabilistic interpretation of linear regression
- Locally weighted linear regression



Locally weighted linear regression
(non-parametric algorithm)

$$w^{(i)} = \exp\left(-\frac{(x^{(i)} - x)^2}{2\tau^2}\right)$$



Logistic regression

Logistic regression

- A supervised learning classification
- For binary classification , target y takes on two values $\{0,1\}$
- Hypothesis (or model)

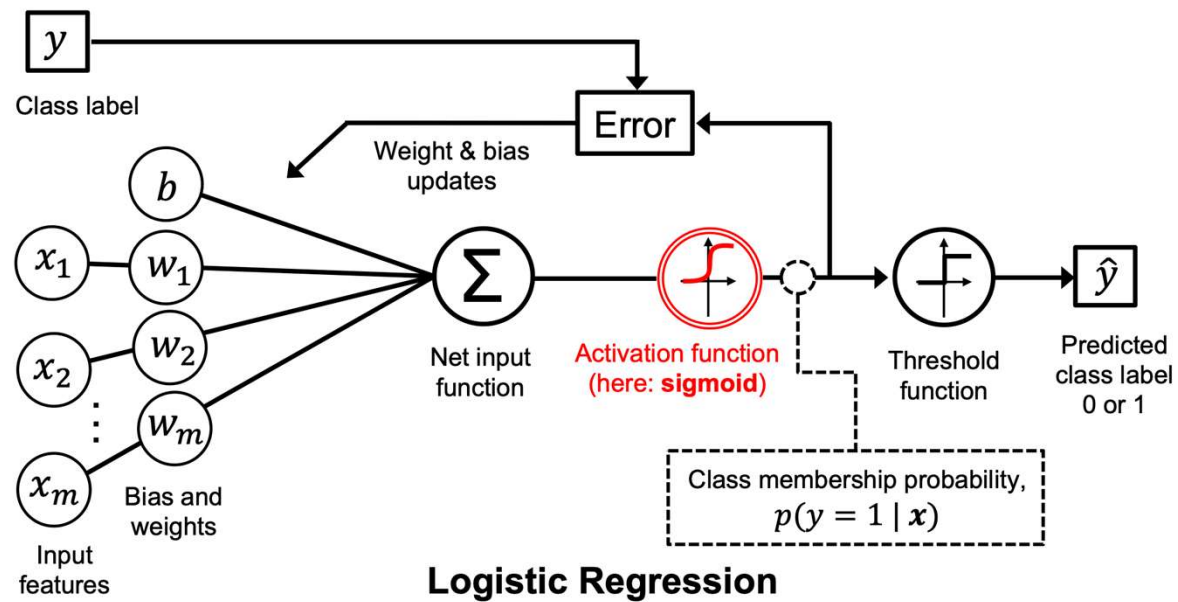
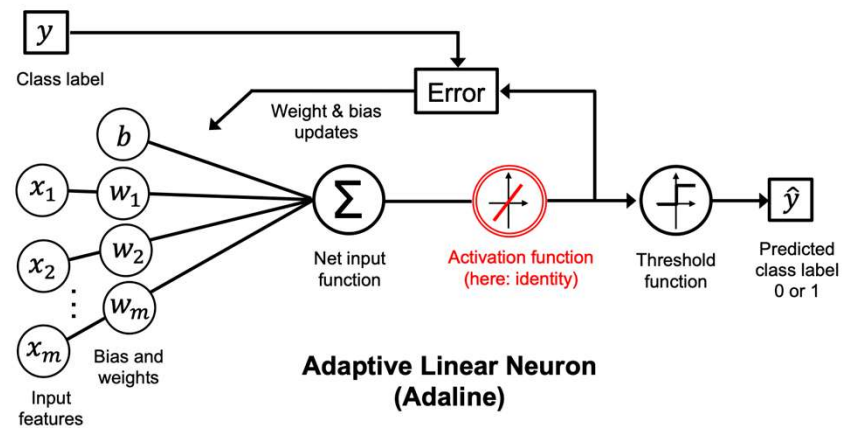
$$h_{\theta}(x) = g(\theta^T x) = \frac{1}{1 + e^{-\theta^T x}}$$

$$g(z) = \frac{1}{1 + e^z}$$

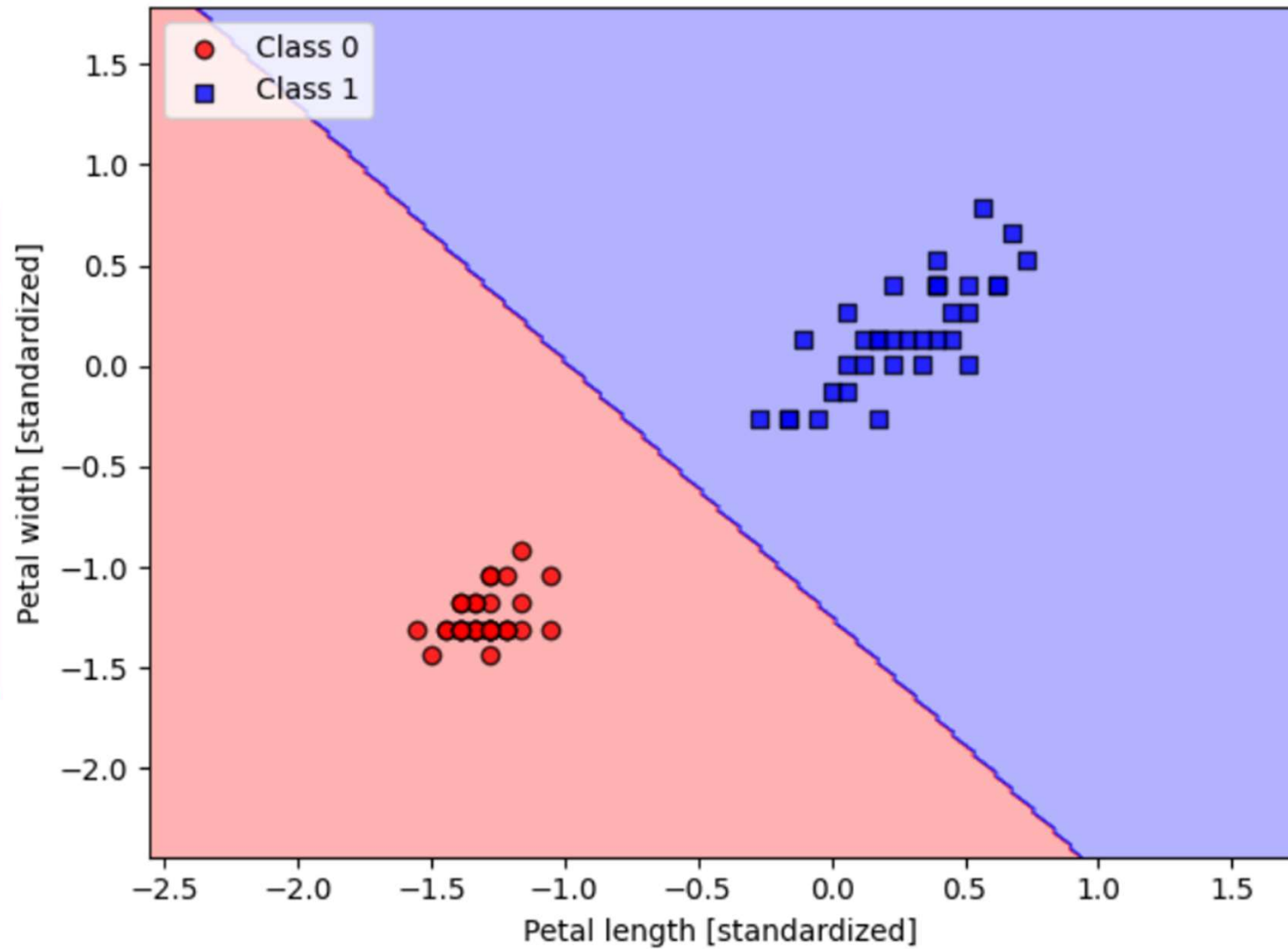


Called
Sigmoid function

Logistic regression compared to Adaline [1]

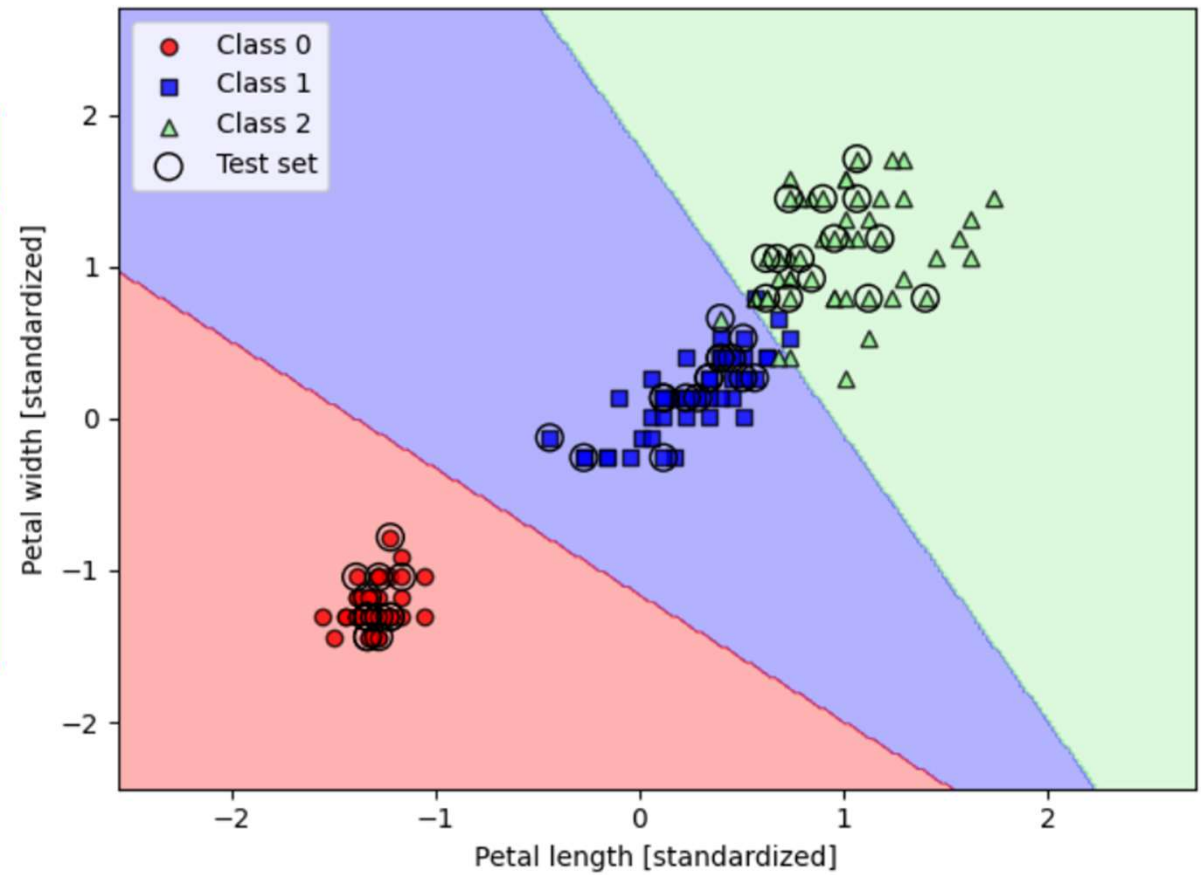


Binary
classification
of the iris
dataset



Multi-class classification

See section 2.3 (p. 24) of [2]



References



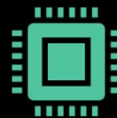
1. S. Raschka, Y. Liu and V. Mirjalili. Machine Learning with PyTorch and Scikit-Learn. Packt Publishing. 2022.

<https://sebastianraschka.com/blog/2022/ml-pytorch-book.html>



2. A. Ng. CS229 Lecture Notes. Stanford University. 2023.

https://cs229.stanford.edu/main_notes.pdf



3. T. Broderick. Introduction to Machine Learning course (6.036). Massachusetts Institute of Technology. 2020.

<https://tamarabroderick.com/ml.html>